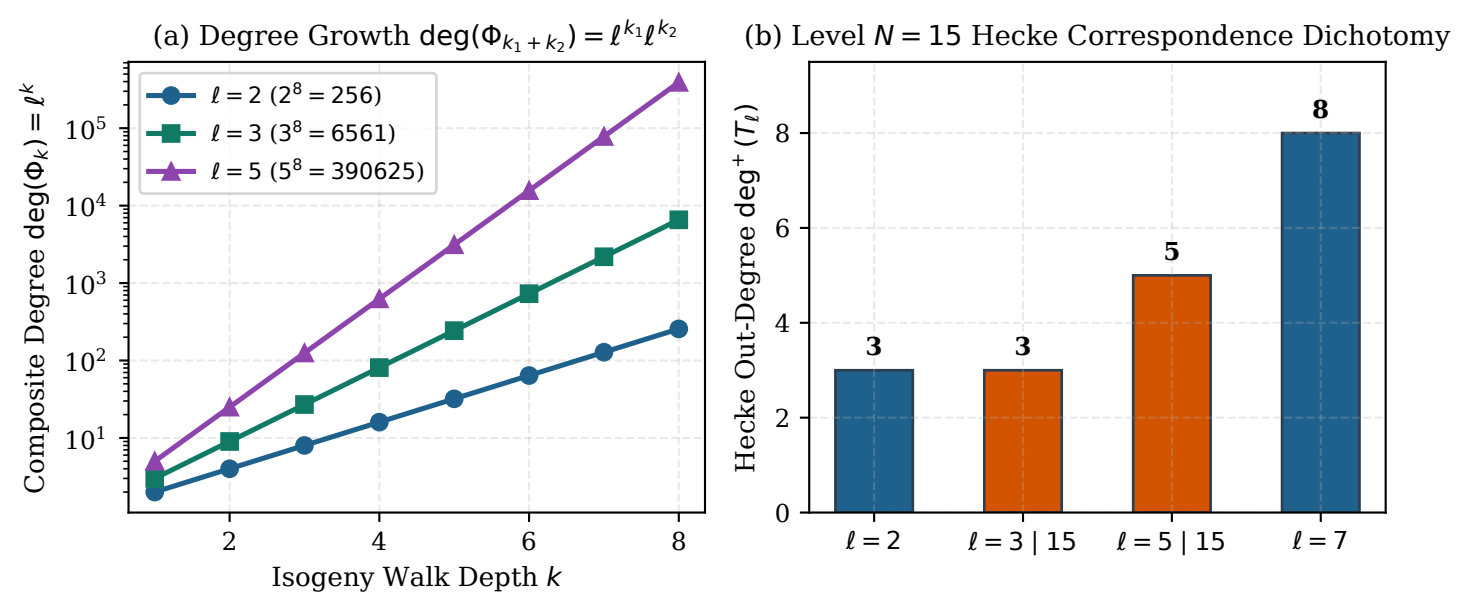
(a) Degree Growth $\deg(\Phi_{k_1+k_2}) = \ell^{k_1}\ell^{k_2}$
$\ell = 2$ ($2^8 = 256$)
$\ell = 3$ ($3^8 = 6561$)
$\ell = 5$ ($5^8 = 390625$)
Composite Degree $\deg(\Phi_k) = \ell^k$
10^1
10^2
10^3
10^4
10^5
2
4
6
8
Isogeny Walk Depth k
(b) Level $N = 15$ Hecke Correspondence Dichotomy
Hecke Out-Degree $\deg^+(T_\ell)$
0
2
4
6
8
3
3
5
8
$\ell = 2$
$\ell = 3 \mid 15$
$\ell = 5 \mid 15$
$\ell = 7$